

2ndLife Revenue OS — Industry Pack System

Companion to: 2ndLife Revenue OS Final Master PRD v4.0

Fixes: the current build hardcodes funeral/insurance terminology ("Funeral Secure Admin," "Policies" tab) instead of deriving the UI from a tenant's selected industry. This document specifies the fix: one engine, N presentation layers.

1. The Core Mechanism

The engine never changes. Only three things change per industry: what things are called, which dashboard widgets show, and which integrations get offered.

At signup, a first-time user picks their industry from a dropdown. That single choice drives an `industry_pack_id` stored on the tenant, and every screen in the app renders itself by reading that pack's config instead of hardcoded strings.

```
// modules/industry-packs/schema.ts
export const industryPacks = pgTable('industry_packs', {
  id: varchar('id', { length: 50 }).primaryKey(), // 'home_services', 'beauty_wellness', 'financial_protection', ...
  displayName: varchar('display_name', { length: 100 }),
  terminology: jsonb('terminology'), // label overrides, see §2
  dashboardWidgets: jsonb('dashboard_widgets'), // ordered widget list + config, see §3
  defaultIntegrations: jsonb('default_integrations'), // suggested integrations, see §6
  complianceRuleset: varchar('compliance_ruleset', { length: 50 }), // NONE | CLINICAL | FINANCIAL | GENERAL
  contentPromptMatrixKey: varchar('content_prompt_matrix_key', { length: 50 }), // maps to Module 3 prompt matrix
  demandKeywordSeeds: jsonb('demand_keyword_seeds'), // starter Demand Radar keywords for this vertical
});

// on tenants table
industryPackId: varchar('industry_pack_id', { length: 50 }).references(() => industryPacks.id),
```

Rendering rule: no component may hardcode a business-facing string like "Policies," "Funeral," "Customers," or "Bookings." Every such label is looked up: `t(tenant.industryPack, 'nav.customers')` → returns "Patients" for clinics, "Guests" for restaurants, "Policyholders" for financial/protection, "Clients" for salons, etc. This is the same pattern as i18n — treat industry variation exactly like language variation.

2. The Industries

#	Pack ID	Industry	"Customer" label	"Booking/Transaction" label	Compliance ruleset
1	home_services	Plumbing, electrical, handyman, pest control, security, pool services	Customer	Job	GENERAL
2	automotive	Mechanics, tyre/fitment centres, car wash, detailing, body shops	Customer	Service Booking	GENERAL
3	beauty_wellness	Salons, barbers, spas, nail bars, aesthetic (non-medical)	Client	Appointment	GENERAL
4	clinical	Dental, medical, physio, chiro, optometry, veterinary	Patient	Appointment	CLINICAL
5	fitness	Gyms, studios, personal training	Member	Class/Session	GENERAL

#	Pack ID	Industry	"Customer" label	"Booking/Transaction" label	Compliance ruleset
6	hospitality	Restaurants, cafés, fast food, catering	Guest	Reservation	GENERAL
7	retail	Retail stores, e-commerce	Customer	Order	GENERAL
8	education	Tutoring, private schools, training providers	Parent/Student	Enrollment	GENERAL
9	pet_services	Grooming, boarding, daycare, training	Pet Owner	Appointment	GENERAL
10	professional_services	Law, accounting, consulting, recruitment, agencies	Client	Engagement	GENERAL
11	construction_property	Contractors, QS, engineering, cleaning, landscaping	Client	Project/Quote	GENERAL
12	financial_protection	Insurance, funeral schemes, financial advisors	Policyholder / Member	Policy	FINANCIAL
13	real_estate	Property sales, rentals, agencies	Client	Listing/Viewing	GENERAL

Compliance ruleset determines what the AI Employee is and isn't allowed to say (§7 of the PRD's guardrails) — CLINICAL blocks diagnosis/treatment claims, FINANCIAL blocks unauthorized advice and restricts offers to pre-approved products, GENERAL just enforces the standard no-invented-facts rule.

3. Onboarding Flow (first-time signup)

1. Sign up (Clerk)
2. "What industry are you in?" – searchable dropdown, the 13 packs above (each with a one-line description + icon, so a plumber doesn't have to guess "Home Services" is them)
3. Business basics: name, locations/branches, services offered, service area/suburbs
4. Industry-specific quick-start questions (pack-specific, see §4 examples):
e.g. clinical → "Do you need HPCSA-compliant messaging?" (pre-toggles CLINICAL guardrails)
e.g. hospitality → "Do you take table reservations or walk-ins only?"
5. Integration prompts – only the ones relevant to their pack (§6), others hidden not just greyed out
6. Dashboard renders with pack-specific widgets, sidebar labels, and starter demand keywords
7. "What should I know today?" AI briefing runs its first pass using pack-specific language

Never show an industry-irrelevant nav item. A plumber never sees "Policies." A funeral scheme never sees "Menu Items." This is done by the dashboardWidgets config on the pack, not by hiding elements client-side after render — the sidebar component takes the pack's widget/nav list as its only data source.

4. Dashboard Widget Config Per Industry

Every pack's dashboard is built from the same widget library (Revenue Command Centre, Leakage Breakdown, Recovery Funnel, AI Briefing, Top Campaigns, Priority Recovery List, Quick Actions) — but each pack chooses which widgets appear, in what order, and with what framing. Below are four fully worked examples; the rest follow the same pattern (§9 gives the template to fill in the remaining nine).

4.1 home_services (Plumber / Electrician / Handyman)

Sidebar: Dashboard · Demand & Content · Jobs · Conversations · Payments · Reviews · Contacts · Reports · Integrations · Settings *(no "Policies" tab)*

Dashboard widgets, in order:

- **Today's Job Pipeline** — new enquiries, jobs in progress, jobs completed today (not "revenue command centre" framing first — trades care about jobs, not abstract revenue language)
- **Emergency Demand Radar** — live surge alerts ("Emergency plumbing demand in Sandton is 2.4× baseline right now")
- **Money Left on the Table** — same leakage engine, relabeled: "Missed calls," "Unanswered WhatsApp," "Quotes gone cold," "No-shows"
- **Recovery Funnel** — relabeled "Job Recovery Funnel": Quotes Sent → Followed Up → Booked → Paid
- **AI Briefing** — "3 missed calls last night got auto-texted, 1 replied and booked for tomorrow 8am"
- **Quick Actions** — Upload customer list, Create emergency campaign, View missed calls, View reports

Primary AI Employee job: instant WhatsApp reply to "emergency plumber [suburb]" leads, missed-call recovery, quote follow-up (stale quotes are the #1 leak in this vertical).

User journey: searches "burst geyser Sandton" spike detected by Demand Radar → owner gets WhatsApp alert to open capacity/raise bids → lead comes in via Google Ad → AI qualifies (what's broken, urgency, address) → quote sent via WhatsApp → if no reply in 2hrs, AI follows up → deposit/booking confirmed → job completed → review request sent → 6-month "time for your annual geyser check" win-back.

Outcome the owner must see: fewer missed calls turning into lost jobs, quotes that used to go cold now converting, and a Rand figure for jobs recovered this month.

4.2 beauty_wellness (Hair Salon / Barber / Spa)

Sidebar: Dashboard · Demand & Content · Appointments · Conversations · Loyalty · Payments · Reviews · Clients · Reports · Integrations · Settings

Dashboard widgets:

- **Today's Chair/Room Utilisation** — booked vs. empty slots today (this vertical's #1 leak is empty capacity, not lead volume)
- **Quiet Hours Radar** — "Tuesday 2–4pm is empty this week — want to send a flash offer to your top 20 clients?"
- **Money Left on the Table** — relabeled: "No-shows," "Unanswered DMs," "Clients who haven't rebooked"
- **Loyalty & Retention Funnel** — Active → At-Risk (30d silent) → Dormant (60d+) → Win-Back Sent → Reactivated
- **AI Briefing** — "4 clients are overdue for their 6-week colour touch-up, 2 DMs on Instagram are unanswered"
- **Quick Actions** — Fill empty slots, Send win-back campaign, Review unanswered DMs, Import client list

Primary AI Employee job: Instagram/Facebook DM and comment response (price/availability questions), booking with stylist preference, rebooking reminders, retail/product replenishment nudges.

User journey: client DMs "how much for balayage" → AI answers with price band, asks hair length/current colour → books with preferred stylist → deposit taken → reminder sequence → post-appointment "book your 8-week touch-up now" → loyalty points accrue → if client goes quiet 45 days, automatic win-back offer.

Outcome the owner must see: empty chair-hours filled, no-show rate down, rebooking rate up, a loyalty balance system clients actually use via WhatsApp.

4.3 hospitality (Restaurant)

Sidebar: Dashboard · Demand & Content · Reservations · Conversations · Loyalty · Payments · Reviews · Guests · Reports · Integrations · Settings

Dashboard widgets:

- **Tonight's Reservations** — covers booked vs. capacity, by service (lunch/dinner)
- **Demand Radar** — local events/weather driving footfall ("Big match at the stadium tonight — expect walk-in surge")
- **Money Left on the Table** — relabeled: "No-shows," "Unanswered menu/delivery questions," "Empty tables Tue–Wed"
- **Guest Retention Funnel** — First Visit → Repeat (2+) → Regular (5+) → Lapsed (60d+) → Win-Back
- **AI Briefing** — "Vegan menu questions are up 40% this week — consider a post," "Table 12 no-show cost ~R850"
- **Quick Actions** — Fill quiet Tuesday, Launch loyalty push, View unanswered comments, Menu update

Primary AI Employee job: table booking via WhatsApp/social, delivery-area and menu FAQ answering, post-meal review requests, loyalty (JOIN/BALANCE/REDEEM), quiet-hour flash offers.

User journey: customer comments "do you deliver to Bryanston?" → micro-reply + WhatsApp CTA → AI confirms delivery zone, sends menu link → books a table or places an order → post-visit review request → loyalty points → quiet Tuesday detected → flash 20%-off lunch offer sent to nearby repeat guests.

Outcome the owner must see: fewer empty tables on slow nights, unanswered social questions down to zero, loyalty members returning more often.

4.4 financial_protection (Insurance / Funeral Scheme — what's currently built)

Sidebar: Dashboard · Demand & Content · Campaigns · Conversations · Payments · Policies · Contacts · Imports · Reports · Integrations · Settings *(this is the only pack where "Policies" belongs)*

This is exactly what's already built in the screenshot — it's correct **for this one vertical only**. The bug is that it's the **only** config that exists right now. Everything else in this document is what needs to sit alongside it, selected at onboarding rather than hardcoded.

Primary AI Employee job: lapsed-policy recovery, premium/renewal reminders, failed-debit-order retry via WhatsApp, pre-approved-offer-only negotiation (FINANCIAL compliance ruleset — no unauthorized advice).

Outcome the owner must see: exactly what's on the current dashboard — recovered revenue verified by webhook, recovery funnel, priority recovery list scored 0–100.

5. Feature Usage by Industry — How the Same Engine Feels Different

Engine (from PRD §3)	home_services	beauty_wellness	hospitality	financial_protection
Demand Radar	Emergency/weather-driven search spikes (burst pipes, load-shedding electrical)	Seasonal/occasion demand ("wedding season updos")	Local events, weather, foot-traffic drivers	Renewal-season and life-event signals (less search-driven, more CSV/date-driven)

Engine (from PRD §3)	home_services	beauty_wellness	hospitality	financial_protection
Content Engine	"How to know you need an emergency plumber" educational posts	Before/after, technique, pricing-transparency posts	Menu features, chef stories, delivery-zone posts	Educational trust content (what your policy actually covers) — heavily compliance-gated
Distribution	Facebook/Instagram/Google Business	Instagram-heavy, TikTok for technique videos	Instagram/Facebook/TikTok, Google Business (reviews matter most here)	LinkedIn/Facebook (lower posting frequency, trust-building tone)
Conversation (WhatsApp)	Job triage, quoting	Booking + stylist preference	Table booking, menu/delivery Q&A	Policy queries, claims-adjacent FAQs (escalate anything complex)
Conversion	Quote → deposit → job	Booking → deposit → appointment	Reservation → (optional pre-order)	Payment/premium collection, not "booking"
Retention	Annual service reminders	Rebooking cycles (colour/cut intervals)	Loyalty visits, birthday offers	Renewal reminders, cross-sell (funeral → life cover)
Recovery	Cold quotes, missed calls	No-shows, dormant clients	No-shows, lapsed regulars	Lapsed policies, failed debit orders (this is the vertical it was originally built for)
Intelligence	Jobs completed, Rand recovered from missed calls	Chair utilisation, rebooking rate	Table utilisation, repeat-guest rate	Recovered premium, policy retention rate

6. Integrations Per Industry

Every pack shares the core integration layer (WhatsApp via Evolution API, Postiz for publishing, a payment provider) but the **optional/suggested integrations offered during onboarding differ**:

Pack	Core (always offered)	Industry-specific (offered, not forced)
home_services	Google Ads, Meta Ads, WhatsApp, Google Business Profile, call tracking (Twilio)	Job/quoting software (e.g. ServiceM8-style CSV import), Google Maps for service-radius
automotive	Google Ads, Meta Ads, WhatsApp, Google Business Profile	Vehicle service-history import, tyre/parts supplier catalogues
beauty_wellness	Instagram/Facebook, WhatsApp, Google Business Profile	Booking calendar (existing salon software import), retail/product catalogue
clinical	WhatsApp, Google Business Profile	Practice management system import (patient records — POPIA-sensitive, encrypted at rest), medical aid claim status feed if available
fitness	Instagram/Facebook, WhatsApp	Membership/billing platform import (for failed debit-order recovery), class-schedule feed
hospitality	Instagram/Facebook/TikTok, WhatsApp, Google Business Profile	POS/reservation system import, menu management (CSV or POS-linked), delivery platform status (Uber Eats/Mr D, read-only)
retail	Meta Ads, Google Ads, WhatsApp	E-commerce platform (Shopify/WooCommerce) product + order feed
education	Meta Ads, WhatsApp	School management system import (enrollment, fee status)
pet_services	Instagram/Facebook, WhatsApp, Google Business Profile	Pet/vaccination record import
professional_services	LinkedIn, WhatsApp, Google Ads	Calendar/scheduling tool, CRM import (existing pipeline)
construction_property	Google Ads, WhatsApp	Quoting/estimation tool import

Pack	Core (always offered)	Industry-specific (offered, not forced)
financial_protection	WhatsApp, Meta Ads	Policy administration system CSV import, payment gateway (debit order status feed)
real_estate	Meta Ads, Instagram, WhatsApp, Google Business Profile	Property listing platform (Property24/Private Property feed)

Integration UI rule: the Integrations page filters to show the pack's core + industry-specific list first, with a "browse all" fallback for anything not on the default list — never present a blank generic marketplace to a first-time plumber and expect them to know they need "Google Business Profile."

7. What Every Industry User Does, Step by Step (universal skeleton)

Regardless of pack, every business owner's actual usage pattern follows the same shape — only the vocabulary and priority order change:

- **Morning:** reads the AI Business Briefing (WhatsApp message + dashboard card) — what happened overnight, what needs a decision today.
- **Reacts to alerts:** demand surge, missed call, unanswered conversation, failed payment — each has a one-tap action ("Target dormant customers," "Review unanswered leads," "Retry failed payment").
- **Approves content:** weekly, reviews AI-drafted posts (industry-appropriate topics from the Content Engine), clicks Approve/Edit/Regenerate — never has to write from scratch.
- **Checks the leakage dashboard:** weekly or monthly, sees the Rand-value leakage report broken down by category, with confidence tags, and the Recovery Score trend.
- **Runs a campaign:** occasionally, launches a targeted recovery/retention campaign (win-back, quiet-hours fill, dormant-customer sweep) with one click from a pre-built segment.
- **Reviews conversations:** occasionally checks in on AI-handled conversations, takes over escalated ones.

8. Success Criteria

8.1 For the AI coding assistant — "done" per feature

A feature is not complete until every row below is verified PASS, in addition to the acceptance criteria already defined in the Execution Plan:

Check	Pass condition
Terminology	Switching a test tenant's <code>industry_pack_id</code> between any two packs changes every user-facing label (nav, widget titles, empty states, AI-generated copy) with zero hardcoded strings remaining — grep the codebase for the old hardcoded terms ("Funeral," "Policies," "Policyholder") outside the <code>financial_protection</code> pack config and confirm zero matches.
Dashboard composition	Each pack renders only its configured widget set, in its configured order — a <code>home_services</code> tenant never sees a "Policies" nav item; a <code>financial_protection</code> tenant never sees "Menu Items."
Onboarding	A new signup, for every one of the 13 packs, completes onboarding and lands on a dashboard with zero placeholder/undefined labels and at least one populated (even if zero-state) widget per configured section.

Check	Pass condition
Compliance gating	A clinical tenant's AI Employee refuses a diagnosis-adjacent question and escalates; a financial_protection tenant's AI Employee never proposes an offer outside the pre-approved list; a home_services/hospitality/other GENERAL tenant has no such restriction incorrectly applied.
Integration offering	The Integrations page for each pack shows exactly its core + industry-specific list from §6 as the default/recommended set, with all other integrations reachable but not default-surfaced.
Data isolation	Switching packs never leaks or reformats another tenant's data — this is a config lookup on render, never a data transformation, and is covered by the existing cross-tenant leak test suite.

8.2 For the business owner — what "this is working" feels like

Regardless of vertical, a paying customer should be able to answer yes to all of these within 30 days of onboarding:

- **"This speaks my language."** Nothing on the dashboard or in an AI-generated message uses the wrong industry's vocabulary — it reads like it was built for their business, not adapted from someone else's.
- **"I can see money, not vanity metrics."** Every dashboard number is in Rand or a directly Rand-adjacent unit (jobs recovered, appointments filled, chairs utilised) — never a bare "engagement" or "reach" figure with no revenue tie-back.
- **"I know what to do next."** The AI Briefing and Quick Actions always point to a specific, one-click next action, not just a report to read.
- **"Nothing falls through the cracks anymore."** Missed calls, unanswered DMs, stale quotes/quotes, and no-shows are visibly being caught and worked, with before/after evidence (e.g. leakage trending down, Recovery Score trending up).
- **"The content sounds like my business, not an AI."** Approved posts reference real things — actual demand signals, actual reviews, actual services — not generic template copy.
- **"I trust the numbers."** Every recovered-revenue figure is one they can trace back to an actual paid, verified transaction — never a projection presented as if it already happened.

9. New Module: LiveSignal — Website Conversation Intelligence

(Naming note: if an earlier build session created a module called `visitor-chat`, this supersedes and evolves it — per the Single Authority Rule, rename/extend that module to `livesignal` rather than creating a parallel one.)

A real-time "live room" layer embedded directly into a client's own website (not a separate site) lets visitors see who else is browsing, ask questions an AI or the business can answer live, and — critically — feeds whatever gets said back into Demand Radar and the Content Engine as first-party demand signal. This is **not a standalone product** — it's a seventh way the engine listens for demand (alongside Google search, social comments, WhatsApp conversations), and it must be configured per Industry Pack exactly like every other module, because "let strangers chat on your website" is right for a restaurant and actively wrong for a funeral home.

9.1 Where it sits in the architecture

LiveSignal is a **Demand Engine input**, not a new engine:

```

LiveSignal (visitor presence + conversation on the client's own site)
↓
Intent/objection/proof classification
↓
Demand Radar (alongside Google/social signals – same 3-question promotion rule)
↓

```

9.2 Data model (Phase A — schema + intelligence layer)

Three entities, tenant-scoped like everything else:

- **live_rooms** — one per embedded room: tenantId, urlPath, mode (community | assist | private), active, peerCount, lastActiveAt. Unique on (tenantId, urlPath).
- **live_signals** — every classified thing said in a room: tenantId, roomId, type (intent | objection | proof | question), category, snippet, confidence, count, consentStatus (na | pending | granted | denied).
- **live_leads** — a visitor whose conversation converted: tenantId, roomId, visitorPhone, question, context (urlPath + intent + objection, structured), converted, leadId.

9.3 Intent taxonomy (deterministic first, AI fallback second)

Classification runs config-driven keyword rules first; only falls through to a tool-grounded AI pass when rule confidence is low (never invents facts, same grounding rule as every other AI surface in the PRD). Categories: price, availability, urgency, financing, value_objection, social_validation, logistics, purchase_intent. E.g. "my geyser is leaking now" → urgency; "has anyone tried this" → social_validation; "do you have payment plans" → financing.

urgency classification triggers the same AI-only, immediate-handoff rule from §9.6/§4.1 — an urgent message never waits in a public room for other visitors to respond; it routes straight to AI intercept.

9.4 Proof detection is consent-gated, not auto-publish

A message showing genuine customer-experience markers ("I've been coming here," "fixed my," "recommend") becomes a proof signal with consentStatus: pending. **Turning a proof moment into a public social post is hard-blocked until the visitor's consent is explicitly granted** — this is a build-time invariant, not a UI nicety, and should be tested as such. Nothing about a real person's words goes public because an owner clicked a button; consent has to exist first.

9.5 Privacy boundary at ingest

The ingest path accepts only already-classified payloads (category, confidence, counts) — never a raw transcript. If a raw transcript field ever arrives, it must be stripped before persistence and logged as a privacy warning, never stored and never returned by any API. This mirrors the PRD's context-minimization rule for AI prompts (§13.2) — the same discipline applies to what LiveSignal is allowed to retain about a conversation.

9.6 Per-Industry Pack configuration (liveRoomMode)

Every Industry Pack config gets a liveRoomMode field, defaulting per pack:

Mode	What it looks like	Which packs default to it
AI-only (no visitor-to-visitor chat)	AI assistant answers live; no public multi-user room	financial_protection, clinical, professional_services
Private/coordinated room	Scoped, invite-only (one family + funeral home; one buyer + agent)	financial_protection (family coordination, opt-in, never public), construction_property, real_estate
Public community room	"N people viewing this" + light moderated public discussion, proof moments surfaced for owner approval	hospitality, beauty_wellness, home_services (general browsing only), automotive, fitness, retail, pet_services, education

home_services is the partial case already noted: general browsing can be public, but any urgency-classified message routes to AI-only mode immediately, regardless of the room's default mode.

9.7 Contextual handoff

When a LiveSignal lead converts, the resulting WhatsApp conversation's **first AI message must reference the actual context** — the question/intent/page the visitor was engaged with — not a generic greeting. This is directly testable: assert the opening message contains the visitor's question topic, not "Hi! How can I help you?"

9.8 Build phasing (explicit scope boundary)

Phase A (in scope now): schema, classifier, proof/consent gate, contextual handoff, Demand Radar/Content Engine feeds, owner-facing UI (rooms + signal feed + proof inbox + leads), Industry Pack LiveRoomMode config. **Phase B** (explicitly out of scope until Phase A is verified green): the actual real-time WebRTC widget and signalling layer that visitors interact with in the browser. Phase A can be fully built, tested, and demoed (including the Sandton Clinic flywheel demo) using synthetic/mock signal ingestion — the real-time transport is a separable, swappable implementation detail (same rule as Postiz/Evolution API elsewhere in this PRD), not a blocker for the intelligence and revenue-loop work.

9.9 Success criteria addition (folds into §8's checklist)

- liveRoomMode is read from the tenant's Industry Pack config, never hardcoded per client, and never defaults to community for financial_protection or clinical — test this explicitly as a compliance gate, not a preference.
- Every classified signal is logged with the same traceability as a social comment and is eligible for the same 3-question → content-brief promotion rule.
- Proof → public content is provably blocked without consentStatus: granted (negative test, not just a UI disabled-state).
- No raw transcript text ever persists in live_signals or is returned by any API (privacy test).
- A detected buying-intent moment produces a WhatsApp handoff whose opening AI message demonstrably references the visitor's actual question/context.

10. Template for Filling In the Remaining Industries

For automotive, fitness, retail, education, pet_services, professional_services, construction_property, and real_estate — not fully detailed above for length — build each pack config using this same worked-example structure:

```
Pack: [id]
Sidebar labels: [nav item list, industry vocabulary]
Dashboard widgets (ordered): [pick from the shared widget library, relabel each]
Primary AI Employee job: [1-2 sentences – what does WhatsApp mostly get used for here?]
Primary leak categories: [which of the 12 universal leaks from the PRD hit this vertical hardest?]
User journey: [discover → enquire → qualify → book/pay → attend → retain → recover, in this vertical's specific language]
Outcome the owner must see: [1-2 sentences, in Rand/vertical-specific terms]
```

Each of these eight follows the same pattern as home_services (trade/service-call verticals: automotive, construction_property), beauty_wellness (appointment/membership verticals: fitness, pet_services), hospitality (footfall/capacity verticals: retail), or financial_protection (pipeline/engagement verticals: professional_services, education, real_estate) — pick the nearest sibling above as a starting template rather than designing from scratch.